Mainboard introduction

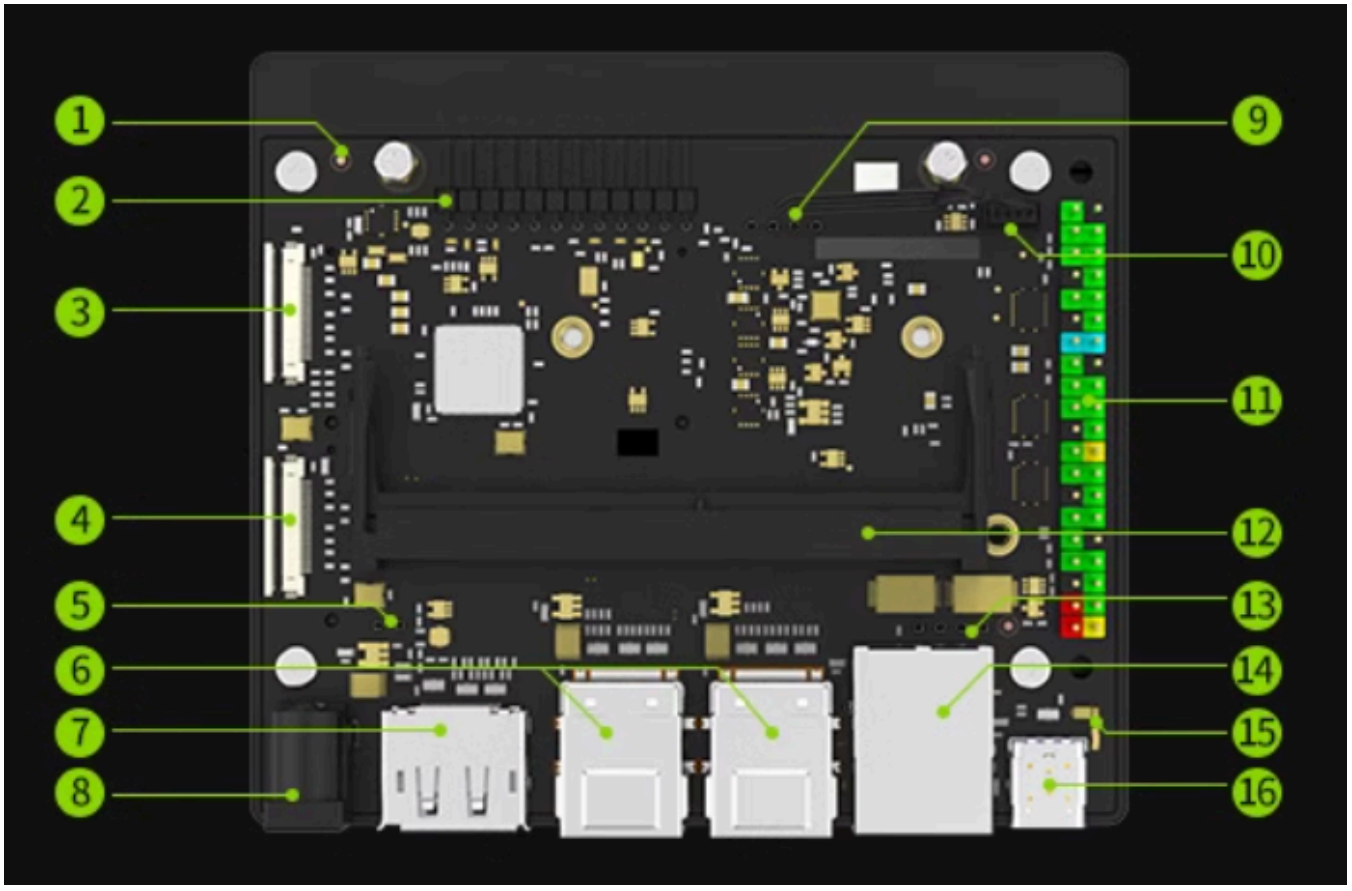
Mainboard introduction

1. Jetson Orin developer kit
    1.1. Front of the board
    1.2. Back of the board
2. Jetson Orin module parameters

1. Jetson Orin developer kit

The main difference between the official Jetson Orin developer kit and the SUB version developer kit: the official kit has no ① power button

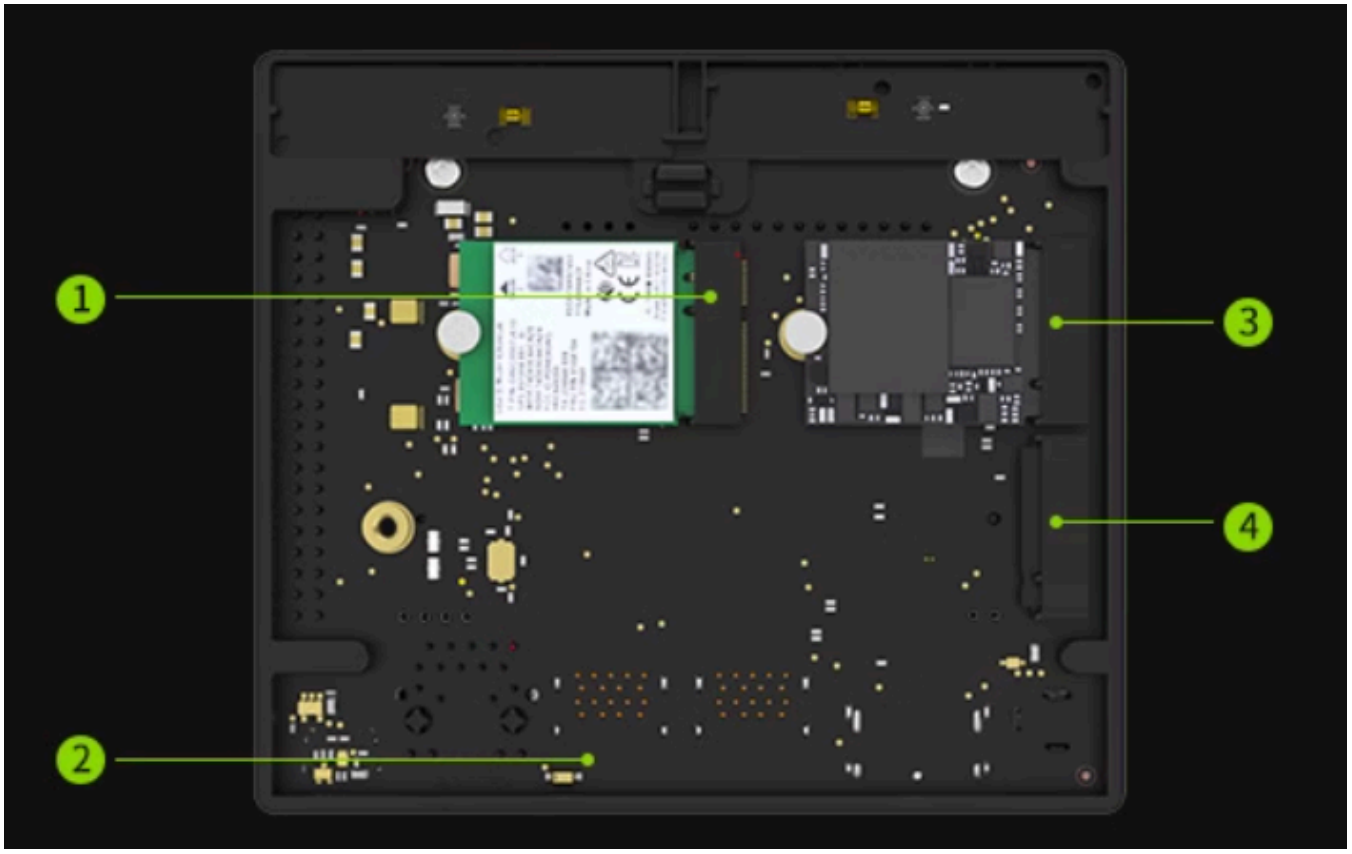
1.1. Front of the board



No.	Description	No.	Description
1	Power button (SUB)	9	CAN bus
2	12-pin button header	10	Fan connector
3	Camera connector 1 (22-pin)	11	40-pin GPIO expansion header
4	Camera connector 2 (22-pin)	12	Core module socket
5	PoE reverse power supply interface (1x2-pin)	13	PoE connector

No.	Description	No.	Description
6	USB 3.1*4	14	Ethernet connector
7	DisplayPort connector	15	Power indicator LED
8	Power connector	16	USB Type C

1.2. Back of the board



No.	Description	No.	Description
1	M.2 Key E connector slot (75 pins)	3	M.2 Key M slot (75 pins)
2	RTC battery holder (optional)	4	M.2 Key M slot (75 pins)

2. Jetson Orin module parameters

	Jetson AGX Orin series				Jetson Orin NX series		Jetson Orin Nano series		
	Jetson AGX Orin Developer Kit	Jetson AGX Orin 64GB	Jetson AGX Orin Industrial	Jetson AGX Orin 32GB	Jetson Orin NX 16GB	Jetson Orin NX 8GB	Jetson Orin Nano Super Developer Kit	Jetson Orin Nano 8GB	Jetson Orin Nano 4GB
AI Performance	275 TOPS		248 TOPs	241 TOPS	157 TOPS	117 TOPS	67 TOPS	67 TOPS	34 TOPS
GPU	2048-core NVIDIA Ampere architecture GPU with 64 Tensor Cores			1792-core NVIDIA Ampere c GPU with 56 Tensor Cores	1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores		1024-core NVIDIA Ampere architecture GPU with 32 Tensor Cores		512-core NVIDIA Ampere architecture GPU with 16 Tensor Cores
GPU Max Frequency	1.3 GHz		1.2 GHz	1.3 GHz	1173MHz	1173MHz	1020MHz	1020MHz	1020MHz
CPU	12-core Arm® Cortex®-A78AE v8.2 64-bit CPU 3MB L2 + 6MB L3			8-core Arm® Cortex®-A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3	8-core Arm® Cortex®-A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3		
CPU Max Frequency	2.2 GHz		2.0 GHz	2.2 GHz	2 GHz		1.7 GHz	1.7 GHz	1.7 GHz
DL Accelerator	2x NVDLA v2					1x NVDLA v2	-		
DLA Max Frequency	1.6 GHz		1.4 GHz		1.23 GHz		-		
Vision Accelerator	1x PVA v2						-		
Safety Cluster Engine	-					-	-		
Memory	64GB 256-bit LPDDR5 204.8GB/s		64GB 256-bit LPDDR5 (+ ECC) 204.8GB/s	32GB 256-bit LPDDR5 204.8GB/s	16GB 128-bit LPDDR5 102.4GB/s	8GB 128-bit LPDDR5 102.4GB/s	8GB 128-bit LPDDR5 102 GB/s	8GB 128-bit LPDDR5 102 GB/s	4GB 64-bit LPDDR5 51 GB/s
Storage	64GB eMMC 5.1				- (Supports external NVMe)		- (SD Card Slot & external NVMe via M.2 Key M)	- (Supports external NVMe)	
Video Encode	2x 4K60 (H.265) 4x 4K30 (H.265) 8x 1080p60 (H.265) 16x 1080p30 (H.265)		1x 4K60 (H.265) 3x 4K30 (H.265) 7x 1080p60 (H.265) 15x 1080p30 (H.265)	1x 4K60 (H.265) 3x 4K30 (H.265) 6x 1080p60 (H.265) 12x 1080p30 (H.265)			1080p30 supported by 1-2 CPU cores		
Video Decode	1x 8K30 (H.265) 3x 4K60 (H.265) 7x 4K30 (H.265) 11x 1080p60 (H.265) 22x 1080p30 (H.265)		1x 8K30 (H.265) 3x 4K60 (H.265) 7x 4K30 (H.265) 11x 1080p60 (H.265) 23x 1080p30 (H.265)	1x 8K30 (H.265) 2x 4K60 (H.265) 4x 4K30 (H.265) 9x 1080p60 (H.265) 18x 1080p30 (H.265)			1x 4K60 (H.265) 2x 4K30 (H.265) 5x 1080p60 (H.265) 11x 1080p30 (H.265)		
CSI Camera	16-lane MIPI CSI-2 connector	Up to 6 cameras (16 via virtual channels) 16 lanes MIPI CSI-2 D-PHY 2.1 (up to 40Gbps) | C-PHY 2.0 (up to 164Gbps)			Up to 4 cameras (8 via virtual channels***) 8 lanes MIPI CSI-2 D-PHY 2.1 (up to 20Gbps)		2x MIPI CSI-2 22-pin Camera Connectors	Up to 4 cameras (8 via virtual channels***) 8 lanes MIPI CSI-2 D-PHY 2.1 (up to 20Gbps)	
PCIe*	x16 PCIe slot supporting x8 PCIe Gen4 M.2 Key M slot with x4 PCIe Gen4 M.2 Key E slot with x1 PCIe Gen4	Up to 2 x8 + 1 x4 + 2 x1 (PCIe Gen4, Root Port, & Endpoint)			1 x4 + 3 x1 (PCIe Gen4, Root Port, & Endpoint)		M.2 Key M slot with x4 PCIe Gen3 M.2 Key M slot with x2 PCIe Gen3 M.2 Key E slot	1 x4 + 3 x1 (PCIe Gen3, Root Port, & Endpoint)	
USB*	USB Type-C connector: 2x USB 3.2 Gen2 USB Type-A connector: 2x USB 3.2 Gen2, 2x USB 3.2 Gen1 USB Micro-B connector:	3x USB 3.2 Gen2 (10 Gbps) 4x USB 2.0			3x USB 3.2 Gen2 (10 Gbps) 3x USB 2.0		USB Type-A Connector: 4x USB 3.2 Gen2 USB Type-C Connector for UFP	3x USB 3.2 Gen2 (10 Gbps) 3x USB 2.0	

	USB 2.0								
Networking*	RJ45 connector with up to 10 GbE	1x GbE 1x 10GbE			1x GbE		1xGbE Connector	1x GbE	
Display	1x DisplayPort 1.4a (+MST) connector	1x 8K60 multi-mode DP 1.4a (+MST)/eDP 1.4a/HDMI 2.1			1x 8K30 multi-mode DP 1.4a (+MST)/eDP 1.4a/HDMI 2.1		1x DisplayPort 1.2 (+MST) connector	1x 4K30 multi-mode DP 1.2 (+MST)/eDP 1.4/HDMI 1.4**	
Other I/O	40-pin header (UART, SPI, I2S, I2C, CAN, PWM, DMIC, GPIO) 12-pin automation header 10-pin audio panel header 10-pin JTAG header 4-pin fan header 2-pin RTC batter backup connector microSD slot DC power jack Power, Force Recovery, and Reset buttons	4x UART, 3x SPI, 4x I2S, 8x I2C, 2x CAN, PWM, DMIC & DSPK, GPIOs			3x UART, 2x SPI, 2x I2S, 4x I2C, 1x CAN, DMIC & DSPK, PWM, GPIOs		40-Pin Expansion Header(UART, SPI, I2S, I2C, GPIO) 12-pin button header 4-pin fan header microSD Slot DC power jack	3x UART, 2x SPI, 2x I2S, 4x I2C, 1x CAN, DMIC & DSPK, PWM, GPIOs	
Power	15W - 60W		15W-75W	15W - 60W	10W - 15W - 25W - 40W	10W - 15W - 25W - 40W	7W - 15W - 25W	7W - 15W - 25W	7W - 10W - 25W
Mechanical	110mm x 110mm x 71.65mm (Height includes feet, carrier board, module, and thermal solution)	100mm x 87mm 699-pin Molex Mirror Mezz Connector Integrated Thermal Transfer Plate			69.6mm x 45mm 260-pin SO-DIMM connector		100 mm x 79 mmx 21 mm (Height includes feet, carrier board, module, and thermal solution)	69.6mm x 45mm 260-pin SO-DIMM connector	

† The Jetson Nano and Jetson Xavier NX modules included as part of the Jetson Nano developer kit and the Jetson Xavier NX developer kit have slots for using microSD cards instead of eMMC as system storage devices.

*USB 3.2, MGBE, and PCIe share UPHY Lanes. See the Product Design Guide for supported UPHY configurations.

**See the Jetson Orin Nano Series Data Sheet for more details on additional compatibility to DP 1.4a and HDMI 2.1

***Virtual Channels for Jetson Orin NX and Jetson Orin Nano are subject to change

Refer to the Software Features section of the latest NVIDIA Jetson Linux Developer Guide for a list of supported features.

View Fewer Specs